

FUNDAMENTALS OF C

C Character set: Every programming languages is having its own character set and by using this character set only we can design the programs [software]. C language is using ASCII character set designed b IBM corporation, which comes with 256 characters. In this we are having 52 alphabets [a-z, A-Z], **10 digits** [0-9], **44 operators** [+,-,...], **14 separators** [, . ; : “ “ () ,....] and remaining all are **special characters**.

Some important ASCII values:

CHARACTERS	ASCII VALUES
a-z	97-122
A-Z	65-90
0-9	48-57
Back space	8
Tab key	9
Enter key	10
Esc	27
+	43
*	42
Space	32

English Language	C Language
-------------------------	-------------------

26 – alphabets	256 – ASCII Characters
English words	Keywords – 32
English sentences	Instructions
English paragraph	Programs
English document	software

C - ASCII – American Standard Code for Information Interchange

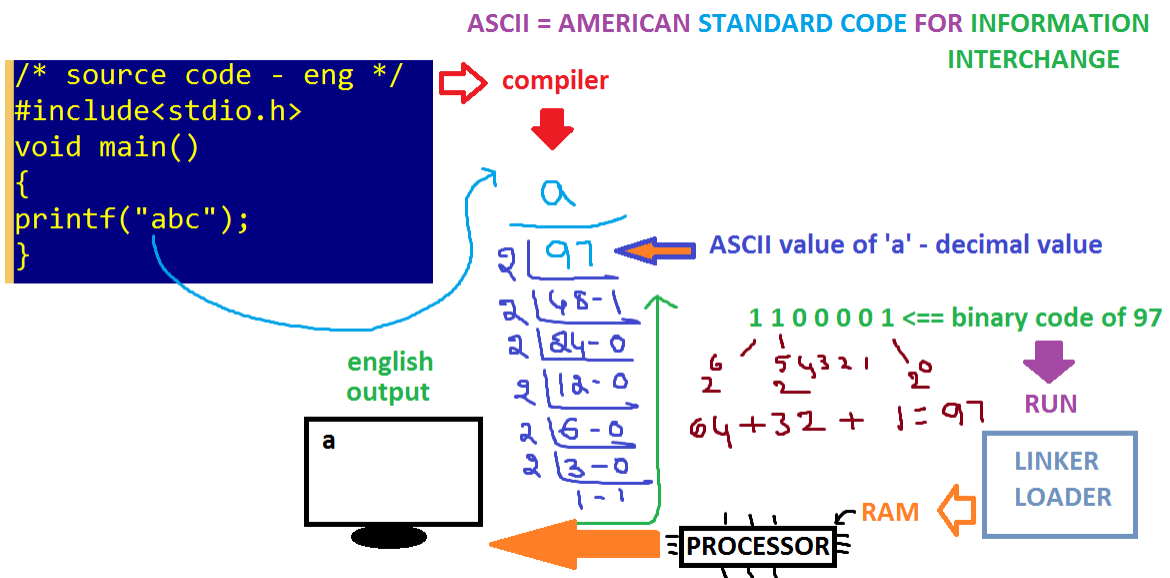
**IBM corporation [International Business Machines]
American company**

Java – UNICODE CHARACTER SET – UNIVERSAL CODE

65536 CHARACTERS

JAVA – WEB APPLICATIONS – WORLD WIDE – DIFFERENT LANGUAGES

COMPILATION AND EXECUTION PROCESS:



Good morning to all. Thank you all.

Good

```
File Edit View Insert Format Tools Help
Mute Start Video Participants Chat New share Pause Share Annotate Slide Control Apps More /watch
Line 10 Col 31 Insert Indent Tab Fill Unindent * E:11AM.C
/* source code - eng */
#include<stdio.h>
void main()
{
char s[100];
int i;
printf("Enter a sentence "); gets(s);
for(i=0; s[i]!='\0';i++)if(s[i]>='A'&& s[i]<='Z')s[i]+=32; /* upper to lower
if(s[i]>='a'&& s[i]<='z') s[i]-=32; /* first letter into upper
printf("U entered %s", s);
}

Enter a sentence sachIN RAMesh tendULKAR
U entered Sachin ramesh tendulkar_
```

A handwritten diagram in blue ink. On the left, the equation $A = 65 + 32$ is written. An arrow curves from the number 65 in this equation to the right, pointing to the equation $a = 97$. Below $a = 97$, the number 32 is written with a minus sign, and a horizontal line is drawn underneath. Below the line, the number 65 is written. A second arrow curves from the number 65 in this subtraction to the left, pointing back to the number 65 in the first equation $A = 65 + 32$.

$$A = 65 + 32 \quad a = 97$$
$$\quad \quad \quad - 32$$
$$\quad \quad \quad \hline \quad \quad \quad 65$$

C TOKENS

The smallest individual words we are using in developing a c program are called C Tokens. They are of different type.